Open Study Buddy

Build your own private AI desk companion that listens, shows, and speaks.



A working Open Study Buddy: Pico 2 W, round display, microphone, and speaker on a breadboard.

KEY FEATURES

- **Listens and speaks.** I2S microphone and speaker with push-to-talk and wake-word detection.
- **Round color display.** A 360 × 360 screen shows a mascot face, text, and live sound meters.
- **Connects to the agent you choose.** Wi-Fi link to a laptop, school server, or cloud service.
- **Push alerts.** School agents can "Push" messages such as “test tomorrow” appear and are spoken aloud.
- **School-aware.** Connect a school calendar so it knows what is due and when the next test is.
- **Complete free textbook.** 18 chapters, runnable MicroPython code, and interactive MicroSims.

KEY BENEFITS

- **Learn by building.** Students learn to write every line of the MicroPython that runs on it.
- **Open and subscription-free.** No vendor cloud and no monthly fee.
- **Privacy by design.** Each person, class, or school decides where the agent runs and what data it may touch.
- **Affordable for a whole class.** About $24 per kit, with no recurring fees.
- **Real engineering skills.** Digital audio, FFT signal processing, networking, and responsible listening.

PARTS & APPROX. COST

RPI Pico 2 W	Microcontroller "brain" with Wi-Fi	**$7**
Display	2.1″ round, 360×360 color	**$7**
Microphone	INMP441 digital I2S	**$4**
Amp	MAX98357A I2S class-D	**$1.50**
Speaker	Small 4–8 Ω speaker	**$3**
Buttons	3 push buttons	**$.50**
Breadboard	Customizable breadboard	**$1.20**
Wires	Dupont jumper wires	**$0.50**
Approximate total		**≈ $24**

Estimated single-unit retail prices in US dollars. Actual prices vary by seller and quantity, and class sets are usually cheaper per kit.

Who it is for: grades 9–12, coding clubs, maker programs, teachers, mentors, and curious hobbyists who already know a little MicroPython.



Read the free book. Build your Buddy.

dmccreary.github.io/open-study-buddy

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Built with MkDocs Material and Claude Code

Scan to open the book